Aashish Dhawan

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SUMMARY

PhD researcher (UF Data Science & Research Lab, Advisor: Dr. Daisy Wang) working on LLMs, multimodal AI, and retrieval systems. **WMT26 and AmericasNLP 2026 shared task winner** (EMNLP + ACL). Published at ACL, EACL, and EMNLP 2026. Built enterprise XAI systems at ExxonMobil on a $168M business.

EDUCATION

Ph.D. in Computer Science — **Aug 2025 – May 2028 (Expected)**
University of Florida · Data Science & Research Lab — Gainesville, FL

- Advisor: Dr. Daisy Wang   Research: Multimodal AI, Low-Resource NLP, Retrieval-Augmented Generation

M.S. in Computer Science — **Jan 2021 – May 2023**
University of Florida — Gainesville, FL

B.Tech. in Computer Science — **Aug 2015 – May 2019**
Kurukshetra University — Haryana, India

PUBLICATIONS

- **Dhawan et al.** "BM25-Augmented Many-Shot Translation for Low-Resource North-Eastern Indian Languages." *WMT26 Indic MT Shared Task @ EMNLP 2026, Budapest.* **[Shared Task Winner]**
- **Dhawan et al.** "Retrieval-Augmented Long-Context Translation for Cultural Image Captioning." *AmericasNLP @ ACL 2026, San Diego.* ACL Anthology. **[1st Place, Overall Shared Task]**
- **Dhawan et al.** "Improving Indigenous Language Machine Translation with Synthetic Data and Language-Specific Preprocessing." *LoResMT @ EACL 2026.* ACL Anthology.
- Driggers-Ellis, . . . , **Dhawan** et al. "MultiScript30k: Multilingual Embeddings for Cross-Script Parallel Data." arXiv:2512.11074. *Under review, ACL ARR.*
- **Dhawan et al.** "Post Processing of Image Segmentation Using CRF." *IEEE INDIACom*, 2019.

EXPERIENCE

Computational Data Scientist Intern — **May 2026 – Aug 2026**
ExxonMobil — Houston, TX

- Built a production ML explainability platform for a **$168M demand forecasting business**, addressing a core challenge: operations teams did not trust AI recommendations. LightGBM models with SHAP, LIME, counterfactual, and drift-detection analysis translate predictions into verified plain-English narratives. Beat the incumbent forecaster by **5.4 pp MAPE** (20.9% to 15.4%) on 970 held-out samples.
- Designed an explainability framework from scratch for LP and MILP optimization decisions (Gurobi, Pyomo/IPOPT), covering which constraints are binding, why an option was not chosen, and how outputs shift when a limit is relaxed. A built-in verification layer only surfaces claims it can prove from the solver output.
- Red-teamed both tools with adversarial inputs to confirm explanations cannot be coerced into fabricating numbers. Built both with a plug-in model interface so they extend to other product lines, teams, and use cases across the organization.

Machine Learning Researcher — **May 2024 – Present**
Data Science & Research Lab, University of Florida — Gainesville, FL

- Built a generalizable **BM25 + many-shot RAG translation pipeline** that won **two shared tasks with the same core architecture**: WMT26 Indic MT (without images, EMNLP 2026) and AmericasNLP 2026 (with Qwen2.5-VL multimodal captioning, ACL 2026), showing that retrieval-augmented few-shot prompting transfers across low-resource language families and modalities.
- Fine-tuned multilingual **mBART-50** with distributed data-parallel (DDP) multi-GPU training, reducing training time by **50%**; chrF++ evaluation across full ablation study published at EACL 2026.
- Engineered a few-shot object detection system for low-resource visual domains (CLIP + SAM + DINO + Adaptive Feature Transfer + OHAC clustering), achieving **15% accuracy gain** over CLIP zero-shot with concept-level visual explanations per prediction.

Adjunct Instructor — **Aug 2023 – Present**
University of Florida — Gainesville, FL

- Designed and delivered Machine Learning and Programming Languages to **300+ students/semester**; led 5+ TAs; mentored capstone projects in LLM chatbots, real-time computer vision, and recommendation systems.

AI Engineer / Consultant — **Aug 2023 – Apr 2024**
Atmosphere Apps — Gainesville, FL

- Engineered GPT-4 and Whisper NLP pipelines for multilingual transcription and prescription recommendation; **95% accuracy** across 10+ languages for **5,000+ active users**.

Graduate Research Assistant **Jan 2021 – May 2023**
Graphics Imaging & Light Measurement Lab, University of Florida Gainesville, FL

- Deep learning pipelines for multispectral image synthesis (ProGAN), neural quantum simulation (**82% compute reduction**), and material scattering prediction via differentiable rendering.

Visiting Researcher **Dec 2019 – Mar 2020**
UBTECH AI Lab, University of Sydney Sydney, Australia

- Multi-source domain adaptation (CNN + GRL adversarial training + moment matching); 71–98% accuracy across 345 classes on a 600k-sample benchmark. Advisor: Prof. Dacheng Tao.

Research Intern **May 2019 – Aug 2019**
Space Applications Center, ISRO Ahmedabad, India

- CNN-CRF satellite segmentation pipeline (LISS-3 + POTSDAM); **648,000× speedup** (36 h → 0.2 s). Published IEEE INDIACom 2019.

SKILLS

Languages: Python, SQL, Java, C++, R
LLMs & Generative AI: GPT-4, LLaMA, Gemini 2.5, Qwen2.5-VL, mBART-50, Fine-tuning (LoRA/PEFT), RAG, LangChain, Prompt Engineering
Multimodal & Vision: CLIP, mCLIP, SAM, DINO, Diffusion Models, Vision-Language Models, Object Detection, Image Segmentation
Search & Retrieval: BM25, FAISS, Dense Retrieval, Vector Search, Semantic Chunking
ML Frameworks: PyTorch (DDP multi-GPU), TensorFlow, Hugging Face Transformers, scikit-learn, LightGBM, XGBoost
Explainable AI: SHAP, LIME, Counterfactual Explanations, Partial Dependence Plots
Tools & MLOps: FastAPI, MCP, AWS, Docker, Git, Gurobi, Streamlit, Pandas, NumPy

SELECTED PROJECTS

Generalizable RAG Translation Pipeline | *BM25, Qwen2.5-VL, Gemini 2.5 Flash, Python* **2025–26**

- One architecture, two competition wins: BM25 retrieval + many-shot LLM prompting won **WMT26 Indic MT** (EMNLP); adding Qwen2.5-VL image captioning won **AmericasNLP 2026** (ACL) for cultural image translation. github.com/dhawan98/mBART50-extended

Explainable Few-Shot Detection for Low-Resource Domains | *PyTorch, CLIP, SAM, DINO* **2025**

- Concept-driven detector for indigenous manuscript glyphs: CLIP + SAM + DINO + Adaptive Feature Transfer + OHAC clustering; **15% accuracy gain** over CLIP zero-shot with hierarchical concept-level explanations served via FastAPI.